

(a) Model-update improvement

Current model version	M1	+2.356 pp
	M2	+0.128 pp
	M3	+1.302 pp
	M4	+1.558 pp
	M5	+0.998 pp
AUC increase (pp)		

(b) Lost improvement (%)

	C0	C1	C2	C3	C4
	-22%				
	-387%	-73%			
	-109%	-49%	-54%		
	-161%	-118%	-121%	-87%	
	-273%	-198%	-216%	-173%	-53%
Cache producer version					

For example, reusing cache produced by M2 in M3 changes AUC by $-54\% \times 1.302 \text{ pp} \approx -0.703 \text{ pp}$.